



Important Formulas: Playing with Constructions

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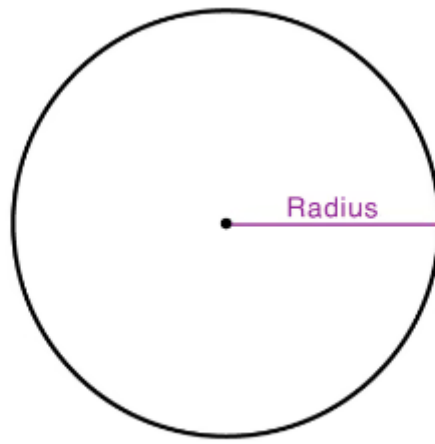
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1. Drawing with Compass

A **compass** is a drawing instrument used to draw circles and arcs with a given radius. It consists of two legs - one with a sharp point that marks the centre, and the other with a pencil that draws the circle.

1. To draw a circle of given radius, mark the centre point and label it **P**.
2. Set the compass to the required distance (for example, **4 cm**) by measuring the gap between the point and the pencil with a ruler.
3. Place the sharp point of the compass at **P** and, keeping the compass fixed at that radius, rotate the compass pencil fully around the point to draw the circle.
4. All points that are **4 cm** away from the centre **P** lie on this circle.
5. The distance from the centre **P** to any point on the circle is called the **radius**. Twice the radius is the **diameter**.



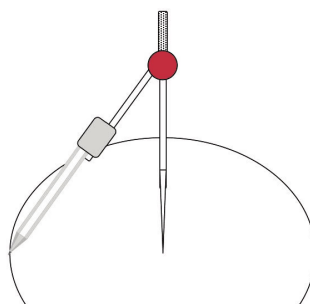
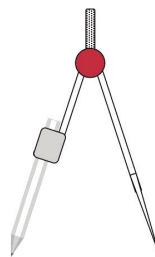
2. Constructing Figures

A compass together with a ruler can be used to construct many figures and decorative patterns. The compass helps copy lengths, draw congruent arcs and semicircles, and transfer measurements accurately.

To draw a simple figure such as a head or a round part of a design, place the compass tip at the chosen centre and draw an arc or circle of the required size.

To make a wavy pattern along a straight line (for example where the central line **AB = 8 cm**), draw a sequence of arcs or half circles with the compass so that each arc meets the next smoothly along the line.

Adjust the compass radius so that the waves are uniform in size and meet neatly along the base line. Use the ruler to mark equal divisions along the base before drawing the arcs to keep spacing consistent.



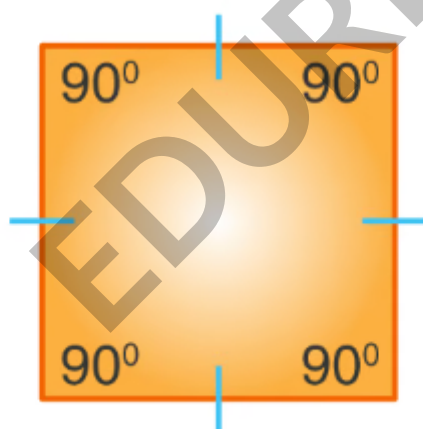
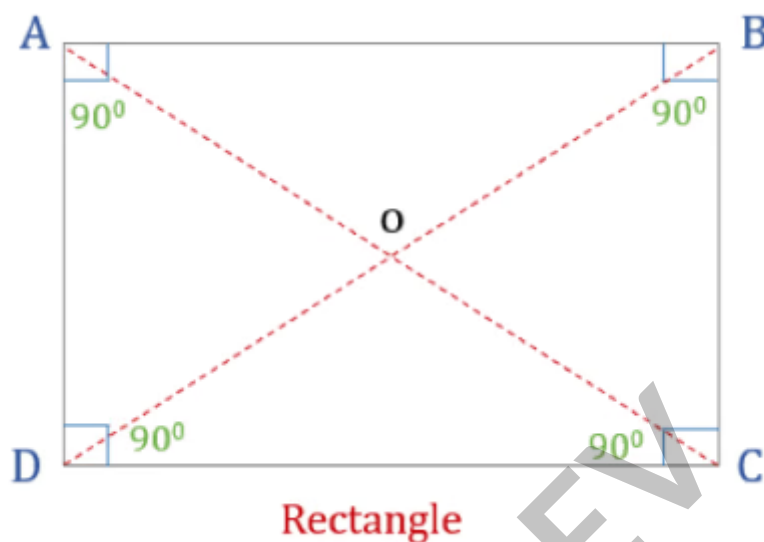
3. Properties of Squares and Rectangles

Squares and rectangles are simple quadrilaterals often used in construction and design problems. Their regular properties are useful when you construct or transform these shapes.

Rectangle: Opposite sides are equal in length; all four interior angles are 90° ; diagonals are equal in length and bisect each other.

Square: All four sides are equal in length; all interior angles are 90° ; diagonals are equal in length, bisect each other at right angles, and are equal in length to each other.

Rotating a square or rectangle does not change these properties: side lengths, right angles and diagonals remain the same after rotation.



4. Constructing Squares and Rectangles

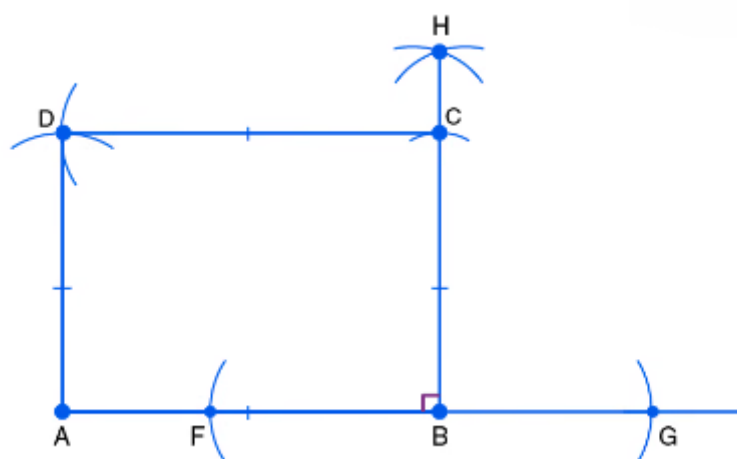
Use a ruler and a compass (or a set square) to construct squares and rectangles accurately. Below are clear, ordered procedures.

To construct a **square** of side 6 cm:

1. Draw a straight line and mark two points A and B such that $AB = 6 \text{ cm}$.
2. At point A construct a perpendicular line to AB.
3. On this perpendicular measure and mark point D such that $AD = 6 \text{ cm}$.
4. With centre B and radius 6 cm draw an arc; with centre D and radius 6 cm draw an arc; their intersection gives the fourth vertex C.
5. Join B to C and C to D to complete the square ABCD, checking that all sides are equal and all angles are 90° .

To construct a **rectangle** with given adjacent sides AB and BC:

1. Draw AB of the required length.
2. At A draw a perpendicular line and mark point D on it so that AD equals the required adjacent side length.
3. From D draw a line parallel to AB; from B draw a line parallel to AD. Their intersection gives C.
4. Join the vertices to complete rectangle ABCD and check that opposite sides are equal and all angles are 90° .



Construction of Square

5. Exploring Distance between Points on Rectangles

When points are placed on or along the sides of a rectangle, distances between them can vary. Understanding minimum and maximum distances helps in optimisation and construction problems.

Place two points X and Y on the sides or edges of the rectangle; the distance XY depends on their positions.

The **minimum distance** between two points on the sides usually occurs when one point is moved to the perpendicular projection of the other onto the same side or when the points are placed closest along adjacent sides.

The **maximum distance** inside a rectangle occurs when the points are at opposite corners; this maximum distance equals the length of the diagonal.

The diagonal of a rectangle with sides of lengths **l** and **b** is given by $\sqrt{l^2 + b^2}$ (Pythagoras' theorem).

6. Breaking Rectangles into Squares

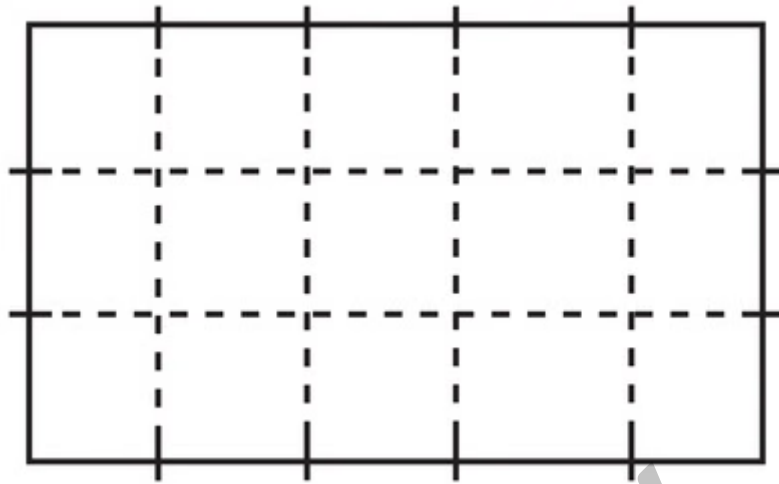
Rectangles can be divided into smaller, identical squares. This is useful for tiling problems and for determining the largest square that fits evenly into a rectangle.

To divide a rectangle of integer side lengths into identical squares, choose a square side that divides both lengths exactly.

The size of the largest such identical square is the greatest common divisor (**gcd**) of the rectangle's side lengths.

Example: For a rectangle 6 cm by 15 cm, the $\text{gcd}(6, 15) = \mathbf{3 \text{ cm}}$. So the rectangle can be divided into squares of side 3 cm.

Continuing the example, number of such 3 cm squares in 6 by 15 rectangle is $(6 \div 3) \times (15 \div 3) = \mathbf{2 \times 5 = 10}$ squares.



These basic constructions and properties form the foundation for many geometry problems. Practise constructing shapes accurately with a compass and ruler, and use properties such as equal sides, right angles and diagonals to reason about distances, tiling and symmetry in geometric figures.